

Combinatorics – 5

Extremal combinatorics

Homework

14 сентября 2026 года

Theoretical minimum. Ramsey: $R(s, t)$ exists and $R(s, t) \leq R(s-1, t) + R(s, t-1) \leq \binom{s+t-2}{s-1}$, while counting colourings gives $R(s, s) > 2^{s/2}$; the infinite version produces an infinite monochromatic set. Turán: a K_{r+1} -free graph has at most $(1 - 1/r)n^2/2$ edges, the extremal example being the balanced complete r -partite graph; for $r = 2$ this is Mantel's bound $n^2/4$. Sperner: an antichain in $2^{[n]}$ has at most $\binom{n}{\lfloor n/2 \rfloor}$ members, by the LYM inequality $\sum_A \binom{n}{|A|}^{-1} \leq 1$ obtained from counting maximal chains. Erdős–Ko–Rado: an intersecting family of k -sets with $k \leq n/2$ has at most $\binom{n-1}{k-1}$ members, by Katona's circle argument. Dilworth: the minimum number of chains covering a finite poset equals the largest antichain; Mirsky is the dual; Erdős–Szekeres on monotone subsequences follows.

1. All the sides and diagonals of a regular 43-gon are coloured red or blue so that every vertex is an endpoint of 20 red and 22 blue segments. A triangle with vertices of the polygon is monochromatic if all three of its sides have the same colour. Suppose there are 2022 blue monochromatic triangles. How many red monochromatic triangles are there? (IMC)
2. Let S be a finite set with n elements and let $\mathcal{F}$ be a family of subsets of S that is upward closed: $A \in \mathcal{F}$ and $A \subseteq B \subseteq S$ imply $B \in \mathcal{F}$. Prove that $f(t) = \sum_{A \in \mathcal{F}} t^{|A|} (1-t)^{|S \setminus A|}$ is non-decreasing on $[0, 1]$. (VJIMC)
3. In a finite set finitely many subsets are chosen so that any two of them meet in at least two elements. Prove that the elements can be coloured in two colours so that no chosen subset is monochromatic. (Mexmat 2007)

4. Let $(X, \preceq)$ be a partially ordered set with the property that whenever $A, B \subseteq X$ satisfy $x \preceq y$ for all $x \in A$ and $y \in B$, there exists $z \in X$ with $x \preceq z$ for all $x \in A$ and $z \preceq y$ for all $y \in B$. (i) Show that every order-preserving $f: X \rightarrow X$ has a fixed point. (ii) Give an example of X and f for which the property holds only for non-empty A, B and f has no fixed point. (ISTCiM 2024)
5. Prove the Bollobás set-pair inequality: if $(A_i, B_i)_{i \leq m}$ are pairs of finite sets with $A_i \cap B_i = \emptyset$ and $A_i \cap B_j \neq \emptyset$ for $i \neq j$, then $\sum_i \binom{|A_i| + |B_i|}{|A_i|}^{-1} \leq 1$. Deduce Sperner's theorem as a special case, and compare the method with the linear-algebraic bounds of Linear algebra – 6.
6. Prove the sunflower lemma of Erdős and Rado: a family of more than $k!(s-1)^k$ sets of size k contains a sunflower with s petals, that is s sets whose pairwise intersections all coincide. Show that the bound cannot be improved below $(s-1)^k$.
7. Prove Mirsky's theorem and use it to show that any sequence of n distinct reals contains a monotone subsequence of length at least $\sqrt{n}$; then construct a sequence showing the bound is sharp.
8. Determine the largest number of subsets of $[n]$ such that no one contains another and no two are disjoint, and compare the answer with Sperner's and with Erdős–Ko–Rado's theorems.
9. In $\mathbb{R}^3$ some n points are coloured. At each step, if four coloured points lie on one line, one may colour any further point of that line. Suppose that in this way every point of $\mathbb{R}^3$ can eventually be coloured, in finitely many steps depending on the point. Find the minimal value of n for which this is possible. (VJIMC)
10. Prove that a family of subsets of $[n]$ in which any two members intersect has at most 2^{n-1} members, that the bound is attained in many ways, and that if in addition no member is contained in another the bound drops to $\binom{n-1}{\lfloor n/2 \rfloor}$.